

Hybrid Solar and Pumped Storage Hydropower System Design in Southern California

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Abstract

In this report, an energy system integrating a community solar farm with pumped hydro storage is proposed for the drought-prone region of East Porterville in southern California. This model presents an opportunity for the region to integrate solar energy into newly developed homes as proposed by the Porterville Community Plan, while additionally addressing the reliability concerns of solar power and drought concerns within the region. The system consists of a 5 MW solar farm of which a portion of energy generation is fed into a 100 KW pump which then pumps water into the nearby Sierra Foothills to be stored in a 100,000 cubic meter reservoir. The water within this reservoir can then flow through a 1 MW turbine into the lower reservoir for power generation during times of peak demand or can be extracted and delivered to the community for agricultural use during droughts. When evaluating the technical production of such a system, it was found that it has the potential to power 702 homes per year and provide a yearly volume of water equivalent to the consumption of 13,500 residents. The LCOE of this system was calculated to be \$0.12/KWh while electricity can be sold for \$0.30/KWh, indicating an overall profitability of the system, with the added benefit of avoiding costs associated with droughts and power outages. Overall, the system presents a net positive opportunity for the town of East Porterville as it ensures a more stable grid and reliable water supply for residents while also increasing the generation of renewable energy in alignment with the town's development goals.

Introduction

As the climate of Southern California grows drier, residents of the town of East Porterville continually face the harshest consequences of this environmental change. Located within the San Joaquin Valley, East Porterville consists of a strong majority of latino farm workers, and is often considered one of the most disadvantaged communities in California (Western City Magazine, 2020). Within East Porterville, a large portion of residents are not connected to a municipal water supply, instead obtaining water through private wells. In a town of approximately 6,700 residents, nearly 1,000 private wells have gone dry over the past few decades (The Guardian, 2020). The lack of a stable water supply in East Porterville has not only threatened the health of residents but has also introduced major financial burdens on the community, as residents undergo costly procedures to obtain water, including importing water or deepening private wells (The Guardian, 2020).



Fig. 1 - Geographic Location of East Porterville

Furthermore, the state of California as a whole currently faces devastating climate change impacts that threaten both human and environmental health. In addition to the frequent and widespread droughts, global warming has introduced dangers including wildfires, ecosystem loss, harm to agriculture, sea level rise, air quality concerns, and more (Bonta, R., 2022). Extreme heat has led to rising electricity demand and equipment failure, putting stress on the power grid and causing frequent outages across the state (Dyson, M., Serrurier, B., & Chaz Teplin., 2020). To combat climate dangers and encourage sustainability, California has turned towards renewable energy, with State Governor Gavin Newsom even recently committing to a path of 100% renewables by 2045 (Governor of California, 2025) . In 2020, California passed a mandate requiring solar photovoltaic systems to be installed along with the construction of nearly all residential and commercial buildings (California Energy Commission, 2020). While these efforts are important for advancing clean energy, the costs associated are often passed onto consumers, introducing yet another disproportionate cost burden onto low income residents in areas like East Porterville.

An energy system integrating solar power with pumped hydro storage presents a unique opportunity to expand renewable energy production in East Porterville while addressing water and energy insecurity concerns. This system would consist of a publicly owned solar farm that mainly generates electricity for residents while diverting a segment of the electricity to pump water into a reservoir for storage. The stored water can then flow into a lower reservoir to generate more electricity in times of peak demand, or can flow to the community for agricultural use during droughts. This model serves to simultaneously expand solar power without shifting costs onto residents, increase grid resilience, and provide a stable backup water supply for the community in times of need. East Porterville is located at the base of the Sierra foothills, which

offer an optimal location for the upper reservoir of the storage system due to their having significant head height and relatively less harsh geographic conditions compared to a typical mountain site. A tertiary wastewater treatment plant is currently under construction in Porterville, and treated water can be drawn from this plant as a method of refill of the reservoir that does not significantly displace an existing water supply (Chavez, M., 2023). During a drought, this tertiary treated wastewater can then be drawn from the reservoir to be used for a wide range of agricultural applications, allowing water that would otherwise be used for agriculture to be redirected to supplement the residential supply.

Health and Safety

Solar panels do not produce any air pollution. The main health concerns for solar farms are heavy metal leaching while they are installed and then end of life toxicity when they are removed or break. One common heavy metal that is used in solar panels is cadmium. Cadmium is known to be highly toxic and carcinogenic, solar manufacturers claim that the risk is minimal due to it being enclosed within a thin film and it is more stable when alloyed into tellurium. However it is important to monitor leaching and prevent breakage that can expose the toxic metals. Another option for photovoltaics is Perovskite solar cells. They have a high solar conversion efficiency of over 22%; however, there are major concerns regarding the dissolved lead content in these materials. If any lead were to leach into the environment, it would be extremely harmful both for ecosystems and for people. There are also risks due to fire and high voltages. For the pumped hydro power system there are different health and safety concerns, which can be broken into two categories: firstly worker health and safety and secondly external work and safety. For worker safety the potential issues are drowning, hydraulic surges, risks

related to big moving machinery, high voltage electricity and pressure shock. For external people the safety concerns are dam and reservoir safety, water quality and biological growths, chemical leakage from machinery oils and other operation chemicals. It is also important to consider natural disasters like earthquakes which are common in California. Another important thing with solar facilities is to maintain proper protocol for keeping workers and anyone who has access to the site safe. This means having solar disconnect switches to shut off connection to high voltages. And making sure that there are no slip, trip or fall hazards near any of the energized infrastructure. This includes regular removal of debris that can build up and cause hazards.

Technical and Economic Analysis

Assumptions and Scope

In the technical analysis of this proposed project, physical assumptions were made to simplify the calculations and to fill gaps in available data. While commercial, industrial, and agricultural energy use may be relevant to this project, the technical and economic analyses of the project primarily use residential energy and water use models due to a lack of available data for other forms of use in the region. Additionally, as a simplifying measure, the geographic scale of this analysis includes only the local area of Porterville, CA. It was found that approximately 890.31 KWh of electricity is used per month per home in Tulare County and electricity can be sold for about \$0.30/KWh (EnergySage, 2024). It was found that the average Porterville resident consumes 69.30 gallons per day (KPCC, 2016). Data for peak electricity demand for the state of California was used to determine frequency of hydropower generation as the demand is not assumed to change much between local areas. Peak demand was found to occur from 4-9 pm

each day (Energy Upgrade California, 2023). Further assumptions regarding technical and economic details of the system are explained along with the calculations below.

Parameter Design Choices

To determine the parameters for the system, a variety of factors were considered involving the desired scale, feasibility, and geographic conditions of the project. A medium sized reservoir capable of storing 100,000 m³ of water was selected to ensure a sufficient amount of water could be stored for the frequent droughts occurring in the region. A head height of 200m was selected to allow significant power to be generated while reducing the difficulties associated with installation and maintenance of a high elevation system. The lower reservoir will be neglected as it does not have a significant effect on system capacity or costs. A 5 MW capacity solar farm was chosen to ensure a substantial amount of power generated for the community while having left over power to pump water into the upper reservoir. To evaluate the amount of energy that could be produced from the solar farm, the National Renewable Energy Laboratory's PV Watts calculator was used (National Renewable Energy Laboratory). After selecting the location of East Porterville, California, the following inputs were entered into the system:

SYSTEM INFO

Modify the inputs below to run the simulation.

DC System Size (kW):	1	
Module Type:	Standard	
Array Type:	Fixed (open rack)	
System Losses (%):	14.08	 Loss Calculator
Tilt (deg):	36	
Azimuth (deg):	180	

Fig.2 - PV Watts Settings

All default inputs were kept with the exception of the tilt angle of the panels, which was set to the latitude of East Porterville. With a 1 kW system, an output of 1,655 kWh/year was obtained, and when scaling this value up to a 5 MW system, the system output 8275 MWh/year. In dividing this produced energy between electricity for homes and power for the pump system, the provision of solar power to homes in the community was prioritized. This was thought to be a better decision as a greater power input to the pump would effectively only result in faster charging of the reservoir, however the reservoir is fairly large and is being consistently gradually charged. In a high-sun region, a 1 MW capacity solar farm is estimated to power around 200-250 homes annually (Energea, 2025). To aim for the powering of at least 500 households, 7500 MWh was designated for household electricity, and the remaining 775 MWh were designated for pump use. This results in a necessary pump capacity of 100 KW to allow for the maximum value of 775 MWh/year to be reached. A pump of this size can be approximated as 80% efficient (Evans, J., 2012). For the turbine, a value of 1 MW was selected to ensure that not too much water is ever being diverted from the tank at once, so there is always a reserve of water which can be sent to the community. The efficiency of a water turbine this size can be approximated to be 90% efficient (Energy Upgrade California, 2023). Since peak demand occurs from approximately 4-9 pm each day, the amount of water diverted from the upper reservoir each day if energy is provided during all peak hours can be calculated as follows, and results in a small fraction of the total water within the reservoir. However, a smaller pump capacity can cause the time for refill of the system to be on the higher end, so it would ultimately be most beneficial to produce hydropower only a fraction of these peak demand times to ensure water remains in the reservoir.

$$Q = \frac{P \cdot \eta}{\rho g h} = \frac{(100000 \text{ W})(.9)}{1000 \frac{\text{kg}}{\text{m}^3} (9.81 \frac{\text{m}}{\text{s}^2}) (200 \text{ m})} = 0.459 \frac{\text{m}^3}{\text{s}}$$

$$0.459 \frac{\text{m}^3}{\text{s}} \cdot \frac{3600 \text{ s}}{\text{hr}} \cdot \frac{5 \text{ hr}}{\text{day}} \simeq 8250 \frac{\text{m}^3}{\text{day}}$$

Power and Water Output

Knowing the typical monthly electricity use of a Porterville household, the number of homes whose electricity needs can be served by the solar farm can be calculated:

$$7500 \text{ MWh} \cdot \frac{1}{.89031 \text{ MWh/month/home} \cdot 12 \text{ month/yr}} \simeq 702 \text{ homes}$$

Based on the 775 MWh of power provided to the pump and its corresponding 80% efficiency, the water storage potential can additionally be calculated by finding the fill flow rate and amount of times per year to fill the reservoir. The time to discharge the water and send it to the community is neglected as discharge works with gravity and is thus likely faster than the recharge time and could also be co-occurring.

$$Q_{fill} = \frac{P \eta}{\rho g h} = \frac{\frac{775 \text{ E6 Wh}}{8760 \text{ hrs/year}} (.8)}{1000 \frac{\text{kg}}{\text{m}^3} (9.81 \frac{\text{m}}{\text{s}^2}) (200 \text{ m})} = 0.0406 \frac{\text{m}^3}{\text{s}}$$

$$V_{fill} = Q_{fill} \cdot \frac{8760 \text{ hrs}}{\text{yr}} \cdot \frac{3600 \text{ s}}{\text{hr}} \cdot \frac{264.172 \text{ gal}}{\text{m}^3} = 3.38 \cdot 10^8 \text{ gallons}$$

Considering that the average Porterville resident consumes 69.30 gallons per day, the quantity of agricultural water which can now be diverted to residents can be calculated:

$$\frac{3.38 \cdot 10^8 \text{ gallons}}{69.30 \text{ gallons/person/day} \cdot 365 \text{ days/yr}} \simeq 13,500 \text{ residents per year}$$

In other words, due to the storage capacity of the reservoir the equivalent of the yearly water use of 13,500 residents which is currently being used for agricultural purposes can now be focused towards these residents.

To determine the yearly power production of the turbine, it is necessary to determine how often the turbine will be operating for each year. The peak demand times for the grid in California occur each day from 4pm-9pm. Since the system is primarily a source of water storage, it is more beneficial for the turbine to provide power when demand is especially high so enough water can remain in the system to be distributed at all times. To account for this, the turbine will only be generating power for 50% of these peak demand times. Thus the calculation for total yearly power output of the turbine is as follows:

$$P = \eta \cdot capacity \cdot hrs/yr = (.90)(1 MW)(\frac{2.5 hrs}{day})(\frac{365 days}{yr}) = 821.25 MWh/yr$$

LCOE

To compute the levelized cost of energy, both the hydropower and solar components of the project must be considered. The major initial costs of the project include the solar farm and the pumped hydropower storage system. The installation cost of a solar farm is approximately \$0.98 per watt (SolarReviews, 2024) and a standard operating cost of \$0.004/KWh will be used. Large hydropower plants range from \$1050-7560/kW to implement (FDE, 2025). Since the plant is being installed at a high elevation, the installation costs are assumed to be in the upper segment of the range of values- \$6000/kW is a reasonable approximation. The operating costs can range from 1-4% of initial installment costs— 3% can be chosen due to the difficulties with high elevation maintenance (FDE, 2025). The lifetime of a solar farm can be assumed to be approximately 25 years, while hydropower can last from 50-100 years (FDE, 2025). A standard discount rate of 5% will be used. Assuming sufficient maintenance, a lifetime of 80 years will be assumed. The LCOE can then be calculated as follows:

Solar:

$$\frac{A}{P} = \frac{0.05(1.05)^{25}}{(1.05)^{25}-1} = 0.071$$

$$ACC_{solar} = \$0.98/W \cdot 5 \cdot 10^6 W \cdot 0.071 = \$347667.04$$

$$OC_{solar} = \$0.004/KWh \cdot (8760h \cdot 5 \cdot 10^3 KW) = \$175200.00$$

Hydro:

$$\frac{A}{P} = \frac{0.05(1.05)^{80}}{(1.05)^{80}-1} = 0.051$$

$$ACC_{hydro} = \$6000/KW \cdot 10^3 KW \cdot 0.051 = \$306177.74$$

$$OC_{hydro} = \$6000/KW \cdot 10^3 KW \cdot 0.03 = \$180000.00$$

Total energy production:

$$821.25 MWh/yr + 7500 MWh/yr = 8321.25 MWh/yr = 8321250 KWh/yr$$

LCOE/profit:

$$LCOE = \frac{ACC_{solar} + OC_{solar} + ACC_{hydro} + OC_{hydro}}{total\ production} = \frac{\$347667.04 + \$175200.00 + \$306177.74 + \$180000.00}{8321250 KWh}$$

$$LCOE \simeq \$0.12/KWh$$

$$profit = revenue - LCOE = \$0.30/KWh - \$0.12/KWh = \$0.18/KWh$$

The revenue received from this system compensates for the capital and operating expenses, thus the system is overall profitable. Additionally, it is estimated that during California's five-year drought, East Porterville spent approximately \$650,000 per month importing water (Western City Magazine, 2020). Power outages can present another costly occurrence for those in California.

Thus the avoided costs from power outages and importing water further contribute to the financial benefit of the pumped hydro storage system, as it can curb the financial impacts of extreme events.

Environmental Protection and Sustainability

The two main considerations for the Porterville and the San Joaquin valley in general are frequent droughts and power outages. They have an average annual precipitation of approximately ~11 inches per year. One of the large concerns with this project is that it will require a large amount of water and end up being a burden on water supply instead of relieving it. To mitigate this issue we will be using tertiary treated waste water effluent to keep the closed loop hydropower system full of water. One concern with using waste water is that it can cause algae blooms or other related issues but with the new tertiary treatment plant that should not be an issue.

Community-Scale Solar Farm

		Scenario A	Scenario B	Scenario C
System Manufacturing & Disposal	Source of PV Modules	Imports from China & Southeast Asia		United States
	Carbon intensity of Module Manufacturing	High-carbon regions in China & Southeast Asia	Average-carbon regions in China & Southeast Asia	Low-carbon U.S. region
	End of Life Treatment	Landfill	Partial recycling	Hypothetical high-recovery recycling
Installation Location Characteristics	Installation Location	Seattle, WA	Fredonia, KS	Phoenix, AZ
	Solar Radiation	Low	Medium	High
	Local Grid Emissions (2022)	Low	High	Medium
	Local Grid Efficiency	High	Low	Medium
	Scenario for Projected Renewable Capacity Cost	Low	Medium	High
Results	Energy Payback Time	1.2 years	0.6 years	0.5 years
	Carbon Payback Time	20 years	2.1 years	0.8 years

Fig.3 United States Solar Data

As seen in the above chart by NREL, solar in Phoenix would pay off its carbon dioxide footprint in under a year. These conditions are extremely optimal and factor in a theoretical high

recovery recycling method however this provides a scale with which to understand the carbon offset. If it took 2-4 years instead of 0.8 it would still be a tremendous carbon benefit as these panels have lifetimes in the 2-3 decades. Additionally solar panels will produce enough energy to offset their production in under a year which makes them extremely sustainable.

Closed-Loop Pumped Storage Hydropower

In comparison to other forms of hydropower, the amount of data related to the environmental impact of closed-loop pumped storage is relatively limited. As of June 2024, there were only 38 operational closed-loop systems globally (PNNL, 2024); therefore, expected impacts from the proposed project are relatively broad and in some cases hypothetical. The construction site choice for the pumped storage hydropower project will play a significant role in contributing to the project's environmental impact. The proposed area for the development, east of Porterville near Lake Success, contains recreational, agricultural, and conservation land use areas. Ideally, the lower reservoir would be constructed on abandoned, open, or nutrient-depleted farmland to reduce negative impact on irrigation and agricultural productivity. In San Joaquin Valley, thousands of acres of farmland are being fallowed every year due to controlled burn bans and cutbacks in irrigation sources by the Sustainable Groundwater Management Act (State Affairs, 2025). Utilizing some of this land for surface water storage as well as for energy storage would arguably create environmental benefits that outweigh the harms caused by development.

Choosing a construction site for the upper reservoir may be more complex, due to aforementioned land use conflicts. However, the foothills near Porterville also contain many abandoned and forfeited mining sites, which currently create safety hazards for hikers and wildlife (National Parks Service, 2022). Very little information is currently available regarding the number and status of these inactive sites, but if a feasible site can be found, constructing the

upper reservoir in an existing open pit or mineshaft could result in lessened environmental impact than if the reservoir were built on previously undeveloped land. The use of former industrial “Brownfield sites” for pumped storage hydropower is an increasingly popular strategy to minimize environmental impact (Weber et. al, 2024). In fact, a large-scale project using inactive mine pits from the former Eagle Mountain Mine has been recently licensed near Desert Center in Riverside County, CA (GEI Consultants). However, abandoned mining sites typically contain some level of hazardous material which must be remediated before construction.

Additionally, the environmental impact of sources chosen to fill the pumped storage reservoirs must be considered. The Eagle Mountain project has been approved to use 3 groundwater sources to fill its reservoirs; however, this decision has faced scrutiny due to drought conditions in the surrounding area (Roth, 2015). Similarly, the most challenging environmental concern for the project proposed in this report would likely be the determination of which water sources to divert for filling reservoirs. The Porterville Area Community Plan, Section 6.5: Water Resources states, “The City is considering purchasing additional surface water from local irrigation and water districts which have water rights on the San Joaquin River and Tule River” (Porterville Area Community Plan, pp. 139). In alignment with Porterville City interests, the San Joaquin River is a possible choice; however, the river is already heavily diverted for agricultural irrigation, filling aqueducts, and municipal use. These diversions have increased the salinity of the river and harmed salmon populations (Walters, 2023). Because of these issues, the San Joaquin River is not an ideal candidate. Further investigation is needed to analyze feasibility of other ground and surface water resources in the regions surrounding Porterville.

The City of Porterville and the Tule River Tribe are currently collaborating to fund a new tertiary wastewater treatment plant to increase the amount of and quality of effluent water that can be diverted for irrigation and other purposes. In a press release, the City of Porterville stated, “The recycled water facility will include the ability for expansion, which will better serve the City’s residents, as the Sustainable Groundwater Management Act (SGMA) limits the City’s use of groundwater both for current use by its residents, as well as to support important future development” (City of Porterville, 2023). Because the system outlined here will also serve as surface water storage for drought relief and be diverted regularly, recycled tertiary wastewater effluent will be regularly used for recharging the reservoirs. As the system is closed-loop and this effluent is already approved for use in agricultural irrigation, using tertiary treated effluent as a water source for this project will not result in water quality impacts that deviate from the current status quo. However, further safeguards could be used to limit water quality degradation. One of these options is reverse osmosis treatment of water held in reservoirs to combat salination, which is a design choice utilized in the Eagle Mountain project (California State Water Resources Control Board, 2020).

Social Sustainability

Porterville Area Community Plan

The economic benefit of this project to community members has been determined partially through consideration of the Porterville Area Community Plan, which is a component of the larger Tulare County General Plan. This document details the Porterville local government’s goals for development of their physical resources, economy, and social services. For this project, the Porterville Area Community Plan serves as a guide for determining whether the proposed

renewable energy system aligns with the interests of the local government of the target area.

Because implementation of a closed loop pumped storage system aligns with several of the stated objectives and policies listed on this document, the project is shown to integrate effectively with the municipality's current development plans.

Implementation Policy 54 of the Open Space and Conservation Element (OSC) of the plan includes, "Support efforts to create additional water storage where needed, in cooperation with federal, State, and local water authorities" (pp. 143). Additionally, the solar farm component of this project aligns with the Energy Resources component of the Porterville plan: "By reducing the amount of energy consumed...and the energy demand per capita plus using more renewable sources of energy, the environment and public health will be better protected" (pp. 153). The implementation of a combined solar and closed-loop pumped storage system is justified through this stated objective because it will decrease energy demand by creating local grid support assets. The solar component will be active throughout the year, and the pumped storage component will generate energy during energy demand peaks in the Porterville Area. OSC Implementation Policy 70 states, "Ensure codes allow for environmentally acceptable alternative forms of energy production and green building techniques" (pp. 153). This policy suggests local government interest in permitting and facilitating the implementation of projects similar to the proposed renewable energy system.

Economic Inequality

The most immediately obvious economic benefit resulting from this project is the creation of new jobs and revenue. The construction of a 6.2 MW solar plant near the Porterville Airport in 2011 created 125 jobs and has produced \$72,792 per year in lease revenue for the city

of Porterville (Schultz, 2011). The proposed 5 MW plant can be expected to create a similar number of jobs and amount of land lease revenue. While specific numbers are difficult to estimate, the pumped storage hydropower aspect of the project could potentially create more than 100 jobs as well. New employment opportunities would stimulate the local economy in the Porterville area. The economic inequality impact of the energy distribution component of the proposed solar and closed-loop pumped storage system is dependent on how the project is integrated into the local power grid. Because energy storage and solar energy utilities could theoretically be distributed differently depending on policy decisions, the role of the solar and pumped storage components of this system in energy equity can be analyzed separately.

Because this project proposes community-scale solar energy generation, the project would have a positive effect on energy equity in Porterville, CA compared to the city's current status quo for energy generation. Although solar panel retrofitting is widely available and incentivized, households in California must pay an average upfront cost of \$15,820 after tax incentives for solar panel installation. This expense makes solar PV installation inaccessible for many homeowners. As of 2020, the California Energy Commission requires all newly constructed single-family and low-rise multi-family housing to be constructed with solar PV installed or with solar PV installation capability (California Energy Commission, 2020). While this responsibility falls on the builder, costs are passed to households through increased construction costs. This estimated price increase for new single family home construction ranges from approximately \$8,400 to \$16,000 (Jackson, 2022). In 2025, the solar PV mandate was extended to all newly constructed nonresidential buildings as well, with additional requirements for battery storage (California Energy Commission, 2025). However, one of the economic incentives for this project is to provide opportunities for households and local businesses to

invest in shares of the community solar farm as an alternative solution for complying with California's Building Energy Efficiency Standards. If homeowners and business owners invest in the proposed community-scale solar farm, the upfront cost to comply with the mandate can be more equitably distributed between members of the Porterville community. This option could lessen financial barriers to building ownership; however, Porterville residents who choose to invest in community solar could see diminished energy cost savings in the long term compared to installing their own solar PV systems.

Additionally, implementation of a community-scale solar farm would benefit community members who are not homeowners or building owners in several ways. As of 2023, the city of Porterville, CA has a 17.9% poverty rate, which is higher than the average of 11.8% for the state of California. The rate of homeownership in Porterville is 54.9% (US Census Bureau, 2022). Homeowners, businesses, and larger entities can directly participate in the community solar farm by investing in shares in exchange for tax credits from the amount of electricity the solar farm generates (McNamara, et. al., 2022). Owners whose buildings predate the 2020 mandate, renters, and other community members are also able to subscribe to community solar and receive tax benefits without owning property. Those who do not directly invest can also experience economic benefits because the electricity cost increase during peak demand times will be reduced if more energy sources are available in the grid (Solar United Neighbors, 2024).

The impact of the energy storage component of this project on economic inequality can be analyzed as well. In California, electric grid blackouts have become increasingly common as wildfires cause damage to infrastructure. Inequity issues arise in these scenarios as economically disadvantaged communities tend to experience more severe energy blackouts, with longer durations and greater frequency. Also, historically redlined communities often have reduced

investment in backup energy generation options (Greenlining, 2024). Through investment in the proposed closed-loop pumped storage hydropower system, the electric grid in the Porterville area will be more independent and have a reduced likelihood of experiencing blackouts (McNamara, et. al., 2022) While Porterville has seen fewer blackouts than cities farther north in California have experienced, the implementation of energy storage in this region would reduce the overall strain on the larger interconnected energy grid and reduce the frequency of power outages in other cities as well. Therefore, the energy grid resilience benefits from the proposed project could have far-reaching impacts in mitigating energy inequity in Southern California.

Engineering Ethics

The ethical analysis of the hybrid solar and pumped storage hydropower system will focus on the “act only in area of competence” canon of engineering ethics. Because the project combines two forms of energy resource design and interfaces with existing electrical and water infrastructure, many specialized skills will be needed to design and implement all components effectively. Firstly, the company or companies hired for this project should have previous experience in designing successful solar, pumped storage, and/or hybrid systems. Because the proposed project has a large number of stakeholders and some complex circumstances, hiring engineers with relevant expertise is important to ensure competence. The companies considered for the project also have the ethical responsibility to turn down the job if their teams do not have adequate experience, skill, or resources to complete the project successfully.

During this project, there is potential for engineers with expertise in only one aspect of the design to be involved with other aspects that require them to use less developed skill sets. One possible remedy for this issue would be to employ a sufficient number of engineers in

multiple disciplines so that all solar components are designed by one team, and all pumped storage components are designed by another team. Other aspects of the project, such as integrating the generation and storage systems together and designing infrastructure for recharging reservoirs with recycled wastewater effluent, could be delegated to separate teams as well depending on the specializations of companies involved in the project.

Additionally, a large amount of permitting and regulatory paperwork would need to be prepared, reviewed, and signed over the course of this project. Federal, state, and local regulations regarding renewable energy standards, water resource management, environmental protection, mining operations, and recycled wastewater effluent use would all need to be followed in order for the project to be approved. Because the project includes multiple components, delegating design tasks to separate, specialized teams and ensuring that all paperwork is prepared by those who worked on the component it concerns would reduce the risk of errors.

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